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RESEARCH

# Validating Pottery Seriation with Radiocarbon Dates: Relative and Absolute Chronology of the Early Bronze Age Necropolis in Mokrin

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## Abstract
Despite the proliferation of radiocarbon dating and the development of different quantitative tools for seriation, the relative chronologies of many prehistoric sites remain untested. These chronologies largely influence current research and should be re-evaluated using more formal methods, when possible. Even when radiocarbon dates are available, they are rarely compared rigorously with relative chronological sequences. In this study, we propose a quantitative method that simultaneously generates and validates seriation solutions with radiocarbon dates. This method accounts for both the complexity of constructing reliable seriations and the inherent uncertainty of radiocarbon measurements and calibration. We demonstrate its application through a case study of the Bronze Age necropolis at Mokrin (Serbia), which is among the most significant Early Bronze Age sites in Southeastern Europe. Using a substantial pottery assemblage from 167 graves, we first classify the ceramic vessels into types based on their shape using geometric morphometrics and then perform a seriation, which we compare with the available radiocarbon dates. Our results indicate that the morphology of the Mokrin funerary ceramics is chronologically sensitive, allowing us to propose a four-phase periodization of the necropolis. In addition to using its material to illustrate the implementation of the proposed method, this research contributes to the region's Bronze Age archaeology by establishing the relative chronology of this important site.

**Keywords** Mokrin necropolis · Bronze Age · Relative chronology · Ceramics · Seriation · Radiocarbon dating

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## Introduction

Since the beginnings of archaeology as a discipline in the early nineteenth century, one of its main tasks has been to establish the chronological ordering of material remains in the archaeological record. Before the development of absolute dating methods, researchers have relied on different relative dating methods, such as seriation or stratigraphic dating (O'Brien & Lyman, 2002). When clear stratigraphy of different archaeological layers was available, it was relatively easy to establish the relative chronological order of the material remains. In other cases, such as establishing regional chronologies based on single-layer sites or the chronological order of graves in a necropolis, the task proved more complex. In such cases, seriation has been employed to construct relative chronological sequences of archaeological entities (Banning, 2020: 309–315; Marquardt, 1978; O'Brien & Lyman, 2002; Porčić, 2018).

As radiocarbon dates became more available, the importance of seriation as a method of relative dating declined. However, seriation remains a valuable tool. It is usually more comprehensive in terms of the sample size, which is particularly important when a large number of entities need to be chronologically ordered (*e.g.*, graves within the necropolis or pits within settlements), and absolute dating is still too costly. Relative chronological sequences can also serve as starting points for creating sampling strategies for absolute dating and/or statistical modeling of absolute (radiocarbon) dates. Finally, seriation inherently relies on cultural transmission (O'Brien & Lyman, 2000; Lipo *et al.*, 1997, 2015; Neiman, 1995; Porčić, 2023), so it can serve to make socio-cultural inferences (*e.g.*, Lipo *et al.*, 1997, 2015).

Despite the proliferation of absolute dating of the material from prehistoric sites during the last several decades, and the development of different quantitative tools for establishing relative chronologies (*e.g.*, Lockyear, 2022; O'Brien & Lyman, 2002; Lipo *et al.*, 2015), the chronology of many prehistoric sites remains poorly known or is based primarily on traditional forms of typological analysis, which were often non-quantitative and more subjective. In this paper, we present a formal quantitative method that simultaneously generates and validates seriation solutions with radiocarbon dates. In order to demonstrate the method, we revisit the question of relative chronology of the Mokrin necropolis, a large and important Early Bronze Age site in Southeast Europe (Fig. 1). The existing relative chronologies of the Mokrin necropolis are based mainly on the traditional typological analysis of ceramics and other material remains (Bona, 1975; Girić, 1971; Gogaltan *et al.*, 2015; Sandor-Chicideanu & Chicideanu, 1989; Soroceanu, 1975; Primas, 1977), with the exception of J. Wagner (2009), who applied a quantitative seriation analysis. We have recently shown that at least two of the previously proposed relative chronologies, including Wagner's, were not supported by radiocarbon dating (Krečković Gavrilović *et al.*, 2025). Even though absolute dates are available for this necropolis, the number of dated graves is still relatively low compared to the total number—only 19 out of 320 graves have published radiocarbon dates (Krečković Gavrilović *et al.*, 2025; O'Shea, 1992). The low number

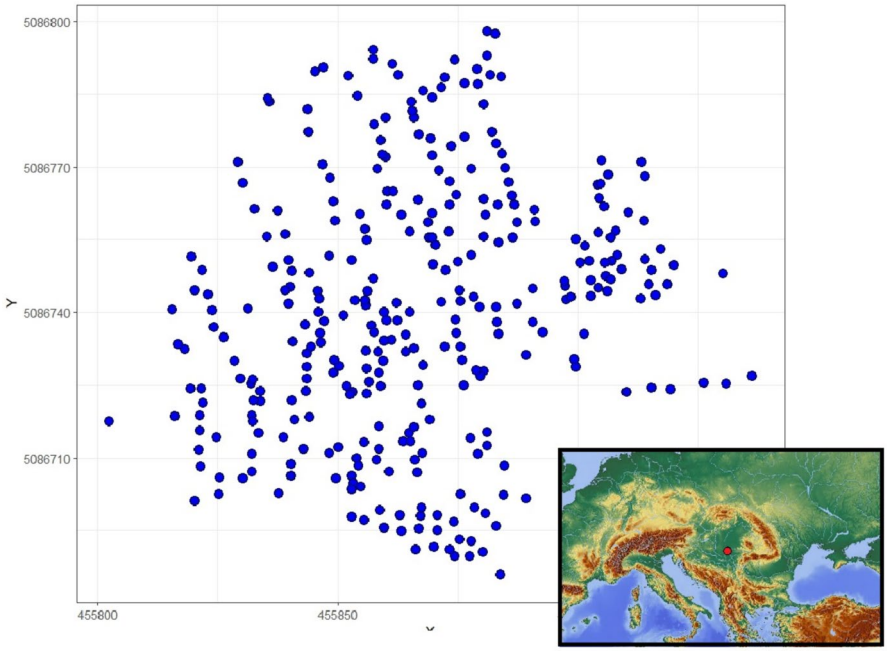


**Fig. 1** Geographical position and plan of the Mokrin necropolis

of dated graves, combined with the fact that all of them have error intervals that are large relative to the duration of the necropolis, precludes the establishment of a comprehensive relative chronology for the site based only on the existing radiocarbon dates. Therefore, it would be useful if a relative chronology could be constructed using another line of evidence. With this goal in mind, we developed and applied a method that combines seriation analysis with available radiocarbon dates to construct and assess the validity of the seriation results.

## Materials and Methods

Mokrin necropolis is situated in Northeastern Serbia, around 13 km north of the town of Kikinda (Fig. 1). It was first excavated in the 1960s and 1970s under the supervision of M. Girić (1971), and excavations resumed in 2020 under the direction of the Faculty of Philosophy, University of Belgrade, and the National Museum of Kikinda (Pendić *et al.*, 2022).

The site is attributed to the Maros cultural group, with distinct burial customs—crouched position, different orientation for males and females—and diverse material culture—ceramic vessels, jewelry from different materials, needles, axes (O'Shea, 1996). The material remains from the site were the subject of many studies (Blagojević, 2020; Krečković Gavrilović, 2022; Krečković Gavrilović & Radinović, 2024; Ljuština *et al.*, 2019; Matić, 2012; O'Shea, 1996; Porčić & Stefanović, 2009;

Radinović & Krečković, 2023; Stefanović & Porčić, 2013; Vitezović, 2017; Žegarac *et al.*, 2021). O'Shea's analysis of grave goods and their spatial distribution across Maros necropolises identified specific artifact types—weapons, hair ornaments, bone needles, and composite sashes—as indicators of social status (O'Shea, 1996). Building on this, S. Stefanović and M. Porčić identified distinct patterns of physical activity among individuals of varying social status at the Mokrin necropolis (Porčić & Stefanović, 2009; Stefanović & Porčić, 2013). However, despite evidence of social ranking and status differences, the overall impression remains that the community buried at Mokrin was largely egalitarian (O'Shea, 1996).

The most recent study of the Mokrin site examined the chronological and spatial development of the necropolis, revealing that it was used continuously between approximately 2100 and 1800 BC, with all areas of the site in use simultaneously (Krečković Gavrilović *et al.*, 2025). Preliminary data from spatial and aDNA analyses suggest that different kinship groups used specific areas of the necropolis simultaneously, indicating designated burial zones based on familial affiliation (Krečković Gavrilović *et al.*, 2025; Žegarac *et al.*, 2021).

Out of 312 graves (Fig. 1) which were excavated at the Mokrin necropolis in the 1960s and 1970s, 222 graves had ceramic vessels as grave goods (around 70% of all graves). In the ongoing excavation campaigns, which started in 2020, 7 new Maros graves were uncovered (3 in 2020, and 4 in 2021) and all of them had ceramic vessels, with the exception of grave 1 from 2020, which was damaged and contained only a couple of long bones. In total, 235 vessels from 167 graves were included in our analysis.

Of the 19 published radiocarbon dates available for the Mokrin necropolis, two are associated with graves lacking pottery (Krečković Gavrilović *et al.* 2025; O'Shea, 1992) and are therefore excluded. Our analysis thus incorporates 17 of the published dates, supplemented by one previously unpublished date (MAMS-77929), presented here for the first time, giving a total of 18 radiocarbon dates available for exploring the correlation between seriation solutions and absolute chronology (all radiocarbon dates are presented in Supplementary file 2 and Supplementary file 4).

## Ceramic Analysis

We recorded the shape of the pottery vessels using morphometric analysis (Radinović & Krečković Gavrilović, 2023). While both form and decoration should be more temporally sensitive than paste and other functional properties (O'Brien & Lyman, 2002: 89), we based our seriation on vessel morphology (shape) rather than ornamentation, as only about 25% of vessels are decorated.

First, we divided the ceramic assemblage into the most common four morphological (quasi-functional) classes: bowls (96), one-handled pitchers (35), two-handled pitchers (63), and amphorae (41) (Fig. 2). In the next step, following Shennan (1997: 288), we measured 10 widths of each vessel at equal intervals from top to bottom, using either vessel drawings or photographs, in order to capture morphological variation. To increase the precision of these measurements, we used the *tpsDig* 2.32 software (Rohlf, 2018) for geometric morphometrics.

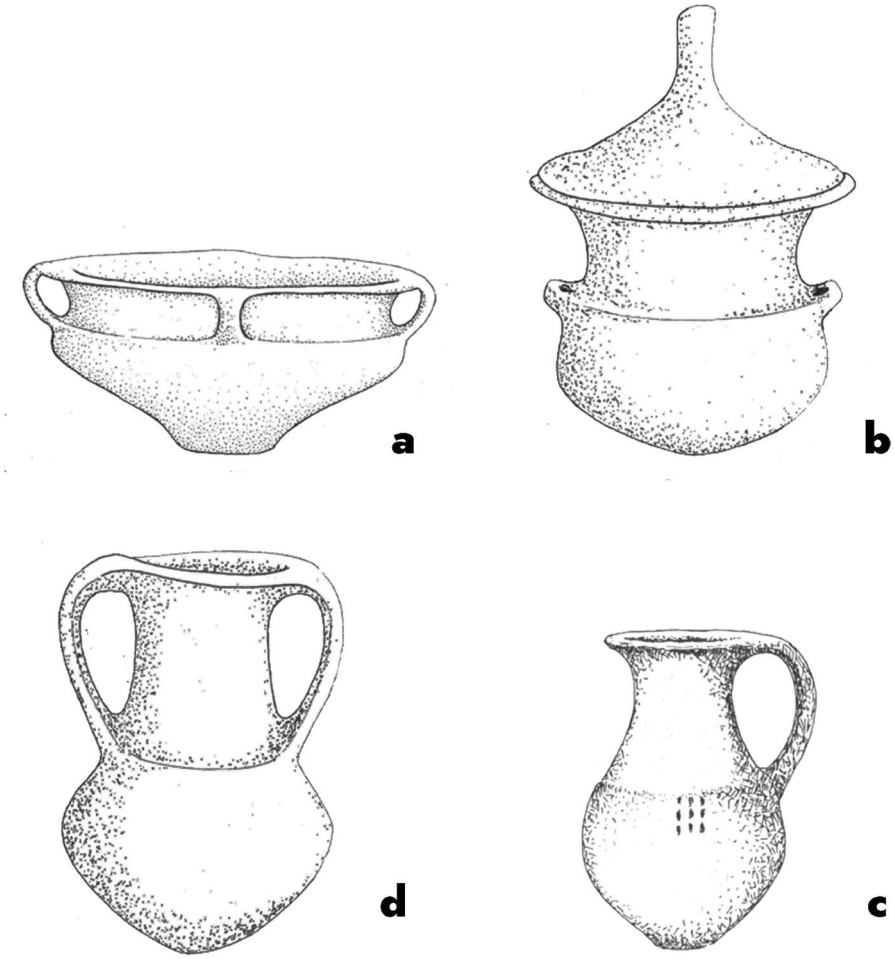


**Fig. 2** An illustration of the major four functional classes of pottery (modified from Girić (1971)): **a** bowl, grave 48; **b** amphora, grave 100; **c** one-handled pitcher, grave 149; and **d** two-handled pitcher, grave 108

To remove the effect of size, the obtained measurements were standardized by dividing them by vessel height. However, as some of the original drawings were clearly done from an angle (the effect of parallax, Macdonald *et al.*, 2020), we subsequently calibrated all standardized measurements using the known ratio between vessel height and rim diameter (Girić, 1971).The calibration was made using a simple formula: dxCal = dDiamCalib × dxRaw/dDiamRaw.

As the visual exploration using PCA showed no clear clustering in the data, the final step for classifying ceramic vessels into types was carried out separately for each quasi-functional class of vessels (bowls, two-handled pitchers, one-handled pitchers, and amphorae). We used hierarchical cluster analysis of Euclidean

distances with Ward's method, which minimizes variance within clusters (*e.g.*, Shennan, 1997: 241–244).

To determine the optimal number of types for each class in our dataset, we used the *NbClust* function in the NbClust package (Charrad *et al.*, 2014). This function evaluates the validity of different cluster numbers using 30 indices and returns the solution that minimizes within-cluster variance while maximizing between-cluster variance. In our case, we set the number of clusters to vary between 2 and 10, and the function identified the most appropriate number of ceramic types within this range.

Finally, we counted the number of ceramic types in each grave. The resulting classification consisted of 13 types: 3 types of bowls, 3 types of one-handled pitchers, 4 types of two-handled pitchers, and 3 types of amphorae.

## Seriation and Validation with Radiocarbon Dates

Our method consists of the following steps:

1. Generating combinations of types. At first glance, it would seem intuitive to use all types and all radiocarbon dates to perform the test. But this is not the best way to proceed, as there is no *a priori* guarantee that all types are constructed to be chronologically sensitive. Moreover, the presence of certain types in graves may be correlated with some other dimensions, such as social status or kinship, rather than time. For this reason, we explore combinations of types in order to find combination(s) which produce such seriation solutions which most likely reflect a chronological signal and which are highly correlated with radiocarbon dates. So, we explore all possible numbers of types (minimum 3 types included in the analysis) and all possible combinations of type subsets, given a certain number of types. For example, if we wish to try out solutions with 5 types, there are.
   $C\begin{pmatrix} 13 \\ 5 \end{pmatrix} = \frac{13!}{5!8!} = 1287$ combinations of 5 types out of 13.
2. Performing correspondence analysis. We used correspondence analysis (CA) as a seriation technique (Baxter, 2003: 204), as it usually outperforms other approaches in producing valid seriation solutions (Lockyear, 2022). For each subset of types, we applied correspondence analysis (CA) to the table of ceramic type counts per grave
3. Rotating the CA axes. Usually, correspondence analysis (CA) is applied as a seriation technique in a straightforward way: scores on the first CA axis are used to construct the relative chronological sequence of the seriated entities. When CA is applied to a dataset that contains a strong temporal signal, *i.e.*, when the frequencies of types follow a unimodal distribution over time, the plot of scores on the first two CA axes often produces a characteristic arch pattern, which is typically taken as an indicator of successful seriation (Banning, 2020: 312–313; Baxter, 2003: 204–207; Neiman & Alcock, 1995; Porčić, 2018).

However, the presence of an arch is neither a sufficient nor a necessary condition for inferring a temporal signal. It occurs when the distribution of types is unimodal, but it may be distorted when the number of types is unbalanced, *i.e.*, when some types change frequency more abruptly than others, or when variation in type frequencies is structured by another factor besides time. For example, the arch may be tilted; in such cases, some authors recommend reading the sequence along the tilted arch (Baxter, 2003: 204–207) rather than projecting points directly onto CA axis 1. A technical way to achieve this is to rotate the arch—that is, to rotate the point coordinates in the two-dimensional space defined by the first two CA axes—and then read the sequence from the projections of the rotated solution onto the *x*-axis.

For these reasons, each solution derived from correspondence analysis was further examined by systematically rotating the two-dimensional CA configuration (the space defined by the first two CA axes) by an angle ranging from 0 to 180° in 5° increments. This procedure allowed us to evaluate multiple possible orientations of the same CA space and identify the one most strongly correlated with time. What the rotation step in our method effectively does is to provide a way of exploring different orderings of individual graves. In principle, one could dispense with correspondence analysis (CA) altogether and simply test all possible permutations of graves. However, this quickly becomes computationally prohibitive as the number of permutations grows dramatically with sample size: once the number of entities to be seriated exceeds 13, the required processing time is measured in years, whereas for more than 20 entities, it is measured in millions and billions of years (Lipo *et al.*, 2015: Table 1). Rotation offers a practical compromise. It begins from a CA configuration, which may already capture aspects of the temporal dimension, since CA is a statistical ordination method, and systematically explores alternative orderings by rotating the axes. Each rotation of the CA solution, *i.e.*, the projections of points onto a rotated axis, is a potential relative chronological sequence (Table 1).

Rotating a correspondence analysis (CA) solution is analogous to rotating principal components in principal component analysis (PCA). The key difference is that in PCA, rotation is performed with respect to an internal criterion—the correlation between variables and principal component axes—whereas in our method, rotation is optimized with respect to an external criterion—the correlation with radiocarbon dates.

4. Validating seriations with radiocarbon dates. Each potential chronological sequence was evaluated against the available absolute chronological evidence in two steps:

   Step 1—drawing calendar dates from calibrated distributions of radiocarbon dates. As radiocarbon dates are probability distributions, rather than point estimates, for each radiocarbon determination, a single calendar date is sampled from its calibrated probability distribution. Calibration was performed with the *rcarbon* package (Bevan & Crema, 2018; Crema & Bevan, 2021) in the *R* programming language (R Core Team, 2024), using the IntCal20 calibration curve (Reimer *et al.*, 2020). Each sampled value represents one possible realization of the absolute

date of a grave. The outcome of this step is therefore a set of simulated absolute dates, consistent with their radiocarbon determinations.

Step 2—calculating correlation between sampled calendar dates and seriation sequence. The absolute value of Spearman's rank correlation coefficient is then calculated between the set of simulated absolute dates and the proposed seriation sequence of graves. Spearman's correlation is appropriate because the relationship between absolute and relative chronology is not assumed to be linear (Lockyear, 2022; Porčić, 2023). Only the absolute value of the coefficient is considered since the direction of correlation is irrelevant: the seriation sequence is arbitrary, and it makes no difference whether it increases or decreases along CA axis 1. The result of this step is one possible correlation value between relative and absolute chronology.

This two-step procedure is repeated 10,000 times (Monte Carlo simulation). The result is a distribution of correlation coefficients that quantifies the degree to which each seriation sequence captures the chronological signal. The same procedure is also used to calculate the statistical significance of each coefficient. If the expected correlation is high, and if most coefficients are both relatively large and statistically significant, we conclude that the proposed relative chronological sequence successfully captures the chronological signal.

As argued in Krečković Gavrilović *et al.* (2025), even if the sequence of graves were perfectly accurate, the correlation between the relative sequence and the associated radiocarbon dates would not necessarily be perfect, due to measurement and calibration errors inherent in radiocarbon dating. To establish a baseline expectation for this hypothetical best-case scenario—that is, the expected correlation between radiocarbon dates and a perfectly accurate seriation sequence, we applied the method developed in Krečković Gavrilović *et al.* (2025).

We generated 10,000 hypothetical best-case scenarios by randomly sampling 18 absolute dates (as it turned out that the best solution is the one which includes all 18 dates, see "Results" section below) from the interval 2077–1822 BC, the most probable temporal boundaries of the Mokrin necropolis based on modeled radiocarbon dates (Krečković Gavrilović *et al.*, 2025). These calendar dates were then back-calibrated to produce radiocarbon measurements, each associated with a standard error randomly sampled without replacement from the set of empirical dates. The Monte Carlo correlation analysis procedure described above was then applied to generate a distribution of correlation coefficients between the relative sequence and the set of simulated radiocarbon dates under conditions of a perfectly accurate sequence.

This provides a frame of reference, enabling direct comparison between the distribution of empirical correlation coefficients and the distribution expected under an ideal, fully accurate chronological sequence.

5. Selecting candidate solutions for relative chronology. To balance accuracy against comprehensiveness, we defined the following criteria that a seriation solution must satisfy in order to be accepted as a candidate solution:

   a. The mean Spearman correlation between (rotated) CA axis 1 and absolute dates must exceed 0.7, while the mean correlation between (rotated) CA axis 2 and absolute dates must remain below 0.3. This ensures that the dominant CA

dimension reflects time, whereas the secondary dimension does not. The correlations are based on the Monte Carlo procedure (described below), which is necessary because radiocarbon dates are represented by probability distributions rather than point estimates.

b. At least 13 radiocarbon-dated graves (out of 18 available) must be included. Reducing the number of types in a seriation analysis inevitably reduces the number of graves included, since many graves contain only a single vessel, which may belong to an excluded type. This may also reduce the number of radiocarbon dates.

c. At least 100 graves (out of 167 available) must be included overall. In some cases, the excluded grave may also be one of those with an associated radiocarbon date, thereby limiting the sample available for validation.

6. Selecting the best solution among candidate solutions. Among the candidate solutions that meet the criteria above, preference is given to the one with the highest mean correlation with radiocarbon dates. However, if another candidate includes at least two additional radiocarbon-dated graves, or more than 20 additional graves overall, it is preferred instead. In such cases, the number of radiocarbon-dated graves carries greater weight than the total number of graves. This procedure ensures that the final choice balances both the breadth of coverage in terms of graves and the strength of chronological support from radiocarbon evidence.
7. Artificial time-averaging of the seriation solution

The chronological position of graves containing a single vessel of the same type cannot be resolved through seriation. This is because such graves occupy identical coordinates in the CA space, making their relative ordering indeterminate. Consequently, seriation cannot provide precise chronological resolution at the level of individual graves.

To improve chronological accuracy at the cost of precision, we aggregated (time-averaged) individual graves into four relatively equally sized groups, according to their rotated CA axis 1 scores, *i.e.*, their positions in the seriation sequence. If the sequence is valid in general, these groups represent more robust temporal phases. The use of equally sized groups reflects the assumption of a constant population through time, an assumption supported by the summed probability distribution of all available radiocarbon dates from the Mokrin necropolis (see Supplementary file 5).

All the quantitative procedures, graphs, and tests were performed in the *R programming language* (R Core Team, 2022) and the following packages: *ca* v. 0.71.1 (Nenadic & Greenacre, 2007), *dplyr* v. 1.1.4 (Wickham *et al.*, 2023), *ggplot2* v. 3.5.1 (Wickham, 2016), *NbClust* v. 3.0.1 (Charrad *et al.*, 2014), *plotrix* v. 3.8–4.8 (Lemon, 2006), *rcarbon* v. 1.5.1 (Crema & Bevan, 2021), *splitstackshape* v. 1.4.8 (Mahto, 2019), and *vegan* 2.6–8.6 (Oksanen *et al.*, 2024). The data and code for reproducing our results are available as supplementary information files.

## Results

There were 5834 candidate seriation solutions that satisfied the conditions of our algorithm (complete list of seriation solutions and their parameters is given in Supplementary file 3 and can be recreated with the R code in the Supplementary files 1 and 2). The mean correlation coefficients between radiocarbon dates and proposed seriation sequences for accepted solutions ranged from 0.7 (the acceptance threshold) to 0.81.

According to our criteria, we selected as the best seriation solution the one with a mean correlation coefficient of 0.75, all 13 types, with all 18 dates, with all 167 graves (all related data are in Supplementary file 4), and with CA axes 1 and 2 rotated by 50°.[1] The solution we selected is the most comprehensive in terms of the number of graves and number of radiocarbon dates among all candidate solutions. Although some solutions achieved higher mean correlations with radiocarbon dates, the differences were marginal when compared to the substantial reduction in the number of graves and radiocarbon dates they included. For example, the solution with the highest mean correlation (Spearman's $\rho = 0.81$) comprised only 115 graves and 13 radiocarbon dates, *i.e.*, 52 graves and 5 dates fewer than our preferred solution. Our reasoning was that comprehensiveness, in terms of both graves and radiocarbon dates, should be given greater weight than small improvements in the mean correlation coefficient.

Moreover, the baseline expectation for the correlation between a hypothetical perfectly accurate seriation sequence and these particular 18 radiocarbon dates is 0.75 (with 95% of correlation coefficients ranging from 0.48 to 0.91). This suggests that the selected seriation sequence achieves a correlation value consistent with what would be expected from a perfectly accurate solution. However, it should be emphasized that this represents a necessary but not a sufficient condition for concluding that this sequence is in fact perfectly accurate. On the contrary, we can be almost certain that it is not, as will be discussed further in the "Discussion" section.

Figure 3 shows the results of the correspondence analysis (with original CA axes 1 and 2 rotated by 50°) on which the seriation of graves is based, and Fig. 4 shows the battleship plot of the selected seriation solution with graves ordered along the first rotated axis of the correspondence analysis. As mentioned above, rotated CA axis 1 has a mean correlation of 0.75 (with a standard deviation of 0.08; 95% of correlation values falling between 0.58 and 0.88) with the radiocarbon dates, and we interpreted it as a temporal dimension, *i.e.*, the sequence of graves along this dimension reflects the relative chronological sequence (Fig. 5).

The correlation between radiocarbon dates and seriation sequence can be visualized by plotting the rotated CA axis 1 scores of the best solution against medians of calibrated radiocarbon distributions (Fig. 6). There is a clear distinction between

[1] The estimated value of the rotation angle is not universal; it depends on the version of the correspondence analysis package used. For instance, two versions of the same software, or two different software packages, may produce different initial (pre-rotation) configurations of points. These configurations are typically mirror images of one another, but the rotation angle that leads to the optimal solution will differ. Therefore, the rotation angle of candidate solutions is meaningful only within the context of a complete run of the code using the same data, on the same computer, and within the same R environment.

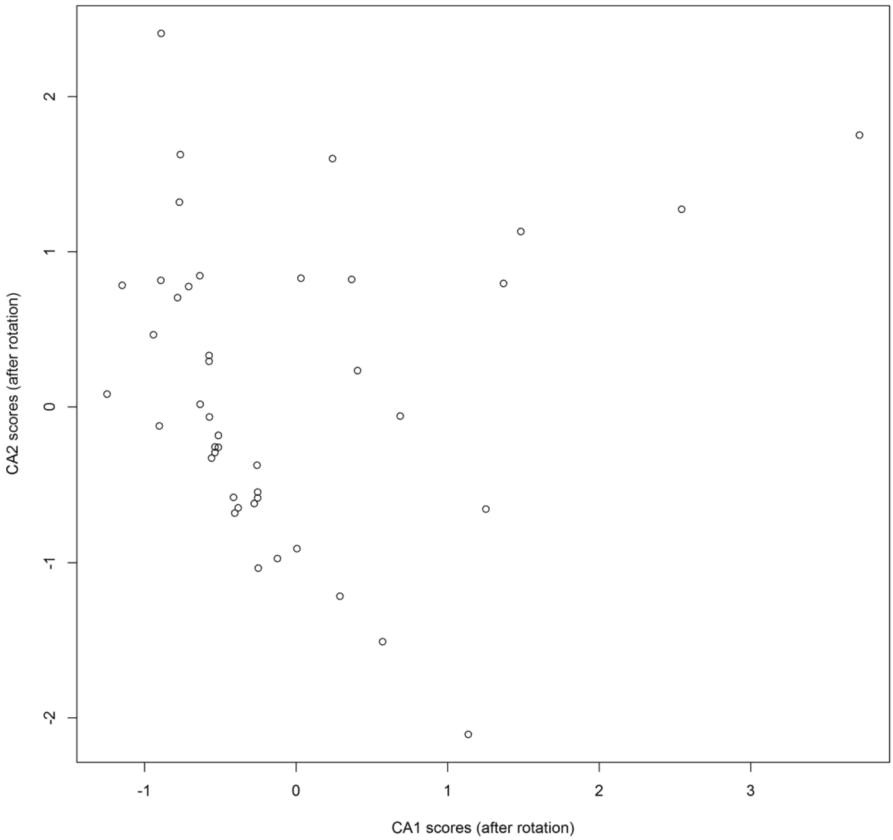


**Fig. 3** Correspondence analysis plot of the best seriation solution showing the position of graves in the CA space defined by the rotated first two CA axes (dimensions) of the best solution

"late" (~1800~1850 median cal BC) and "early" (~2100~1900 median cal BC) graves. It should be kept in mind, though, that median values of calibrated dates are just one potential realization of their distributions, so their positions on the *y* axis may not reflect the true calendar date of a grave, but rather its expected value. Likewise, rotated CA axis 1 scores should be interpreted as an ordinal variable: since they are derived from the chi-squared metric, distances between points in the CA space cannot be read as Euclidean distances based on interval or scale variables.

Our best seriation solution includes 167 graves. We divided them into four chronological phases, with approximately an equal number of graves per phase (47, 49, 30, and 41 graves per phases 1–4, respectively). The division into phases was made by clustering adjacent graves along the sequence of rotated CA axis 1 scores. The relative frequencies of types across this four-phase sequence display a characteristic unimodal pattern (Fig. 7), which typically arises when the sequence reflects temporal ordering (O'Brien & Lyman, 2002; Porčić, 2013; but see Porčić, 2023: Chapter 5). However, as argued by Porčić (2023: 166), high unimodality is a necessary but not sufficient condition for accurate seriation, and the inferential limits of this

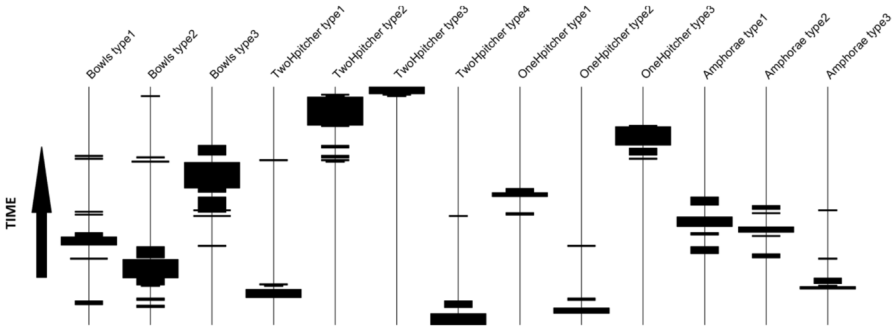


**Fig. 4** Battleship plot of the best seriation solution with the sequence of graves along the rotated CA 1 axis

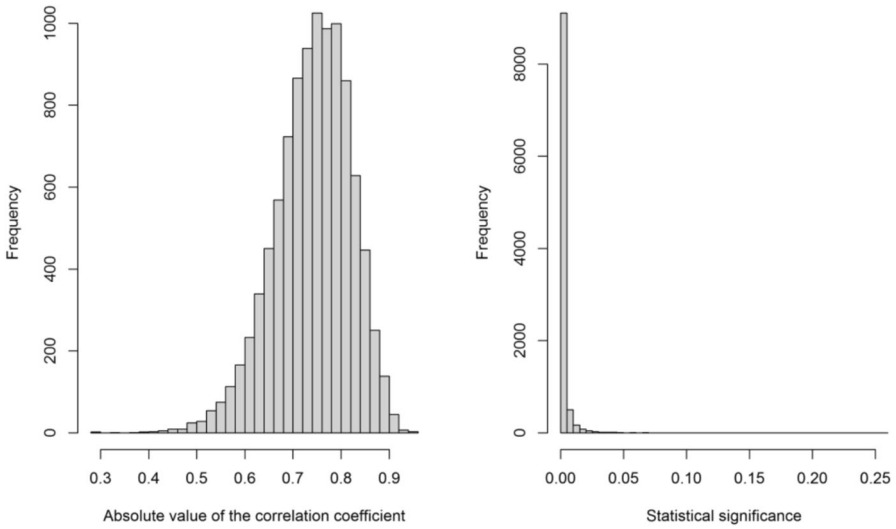


**Fig. 5** (Left) Distribution of the values of Spearman's correlation coefficient between absolute dates and rotated CA axis 1. (Right) Distribution of associated statistical significance (*p*) values

result must be acknowledged. The aggregation into phases also rests on the assumption that both the mortality regime and the population size remained relatively constant over the duration of the cemetery's use.

## Discussion

Although the results of our seriation appear satisfactory at first glance, several important caveats must be acknowledged. It should be noted (Fig. 4) that several chronological sections are represented with graves which only have one class of

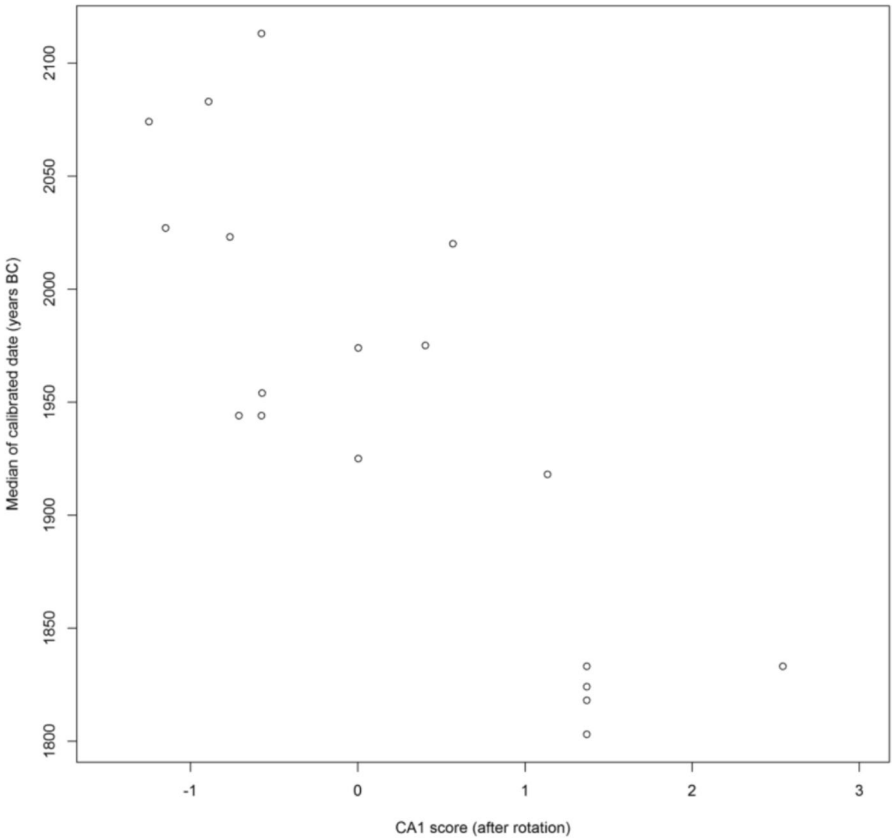


**Fig. 6** Rotated CA axis 1 scores plotted against medians of calibrated radiocarbon distributions. Grave numbers are shown in the plot

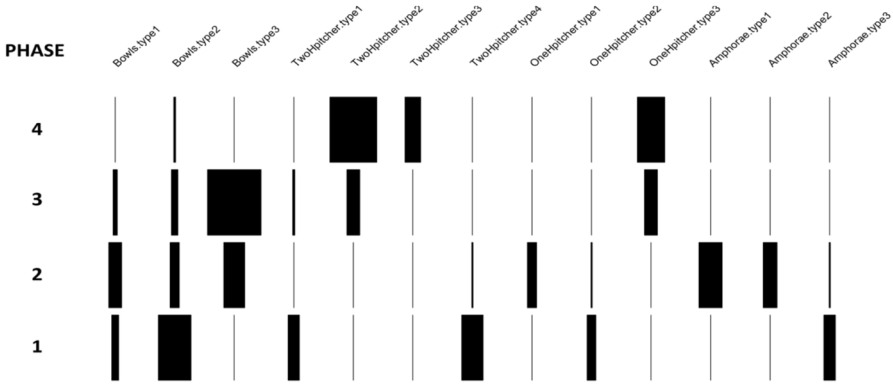


**Fig. 7** Battleship plot of relative type frequencies in 4 phases derived by aggregation (time-averaging) of graves into 4 groups along the sequence defined by the rotated CA axis 1 of the best seriation solution

vessels (*e.g.*, bowls or pitchers). This is more likely to be the artifact of the method than a reflection of the funerary practice—seriation of graves which usually have one vessel per grave is challenging, as any algorithm will struggle to make connections between such graves. Such connections are usually established via graves with multiple vessels. However, these are sparse and in reality there is often no sufficient information to accurately place an assemblage in the right place in the sequence, but the solution is still produced by statistical forcing (*e.g.*, correspondence analysis will always produce a result). In addition to this, there are many graves which only have a single vessel of the same type. These graves all have the same score on the rotated CA 1 axis suggesting that they are contemporary, which is not often the case (*e.g.*, it is not likely that two individuals were interred at exactly the same time). Aggregation of graves into phases partially solves this problem—it increases the accuracy at the cost of reduced precision.

When the inferred four chronological phases are projected spatially (Fig. 8), we can see that there is no spatial clustering. Graves from all phases are found throughout the necropolis, consistent with the conclusion of Krečković Gavrilović *et al.* (2025), based on the spatial distribution of radiocarbon dates, that all areas were used continuously. This pattern also aligns with evidence of kinship organization (Žegarac *et al.*, 2021), suggesting that different families or kin groups occupied distinct parts of the necropolis over extended periods.

When distributions of medians of calibrated dates are shown as boxplots for each phase (Fig. 9), we can see that the four-phase model is generally consistent with the radiocarbon data, but it is not perfect—phases 2 and 3 do not seem to be well resolved. This may indicate that these two phases are not well defined (*e.g.*, that graves from these phases come from a single phase), or what is more likely, they may reflect the shape of the radiocarbon curve—the graves may indeed belong to different phases, but radiocarbon measurements from 2000 to 1950 BC are similar, as the calibration curve has a small wiggle in this interval. This explains why, even in an ideal hypothetical scenario where the seriation sequence is perfectly accurate and precise, the expected correlation between calendar dates and the seriation sequence is 0.75 rather than approaching 1.

Therefore, the aggregation of graves into phases may not fully resolve the problem of inaccurate positioning of individual graves in the seriation sequence. It will only reduce the problem, but it is still possible that some of the graves are assigned to the wrong phase.

To test the sensitivity and stability of our best solution to perturbations in the data, we randomly removed a subset of graves, restricted to those without radiocarbon dates, to assess how such exclusions would affect the validity of the seriation sequence, that is, its correlation with the radiocarbon dates. To do so, we performed the following additional analysis (R code in Supplementary file 6):

1) We randomly removed $n$ graves without radiocarbon dates from the full dataset, with $n$ ranging from 1 to 67; for each $n$, we performed 100 random removals.
2) For each reduced dataset, we performed CA and calculated correlation coefficients between the seriation sequence and the medians of calibrated radiocarbon dates

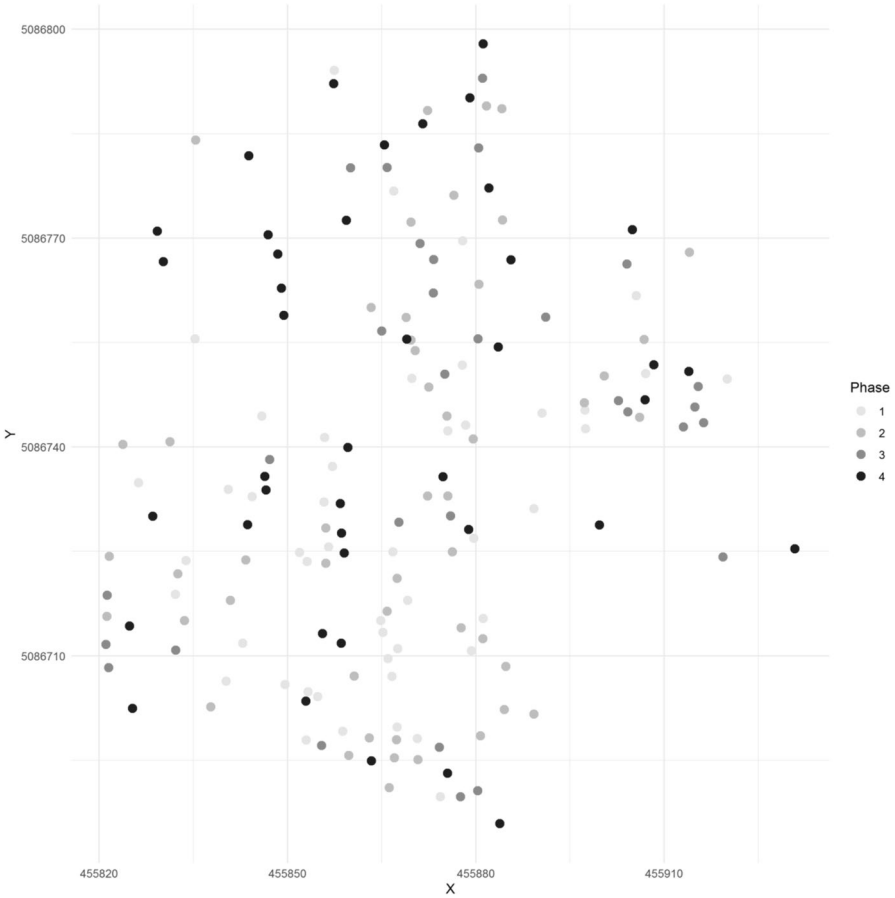


**Fig. 8** The spatial distribution of seriated graves with phase membership indicated by grey scale

across all possible rotations, retaining the maximum correlation for each case as a measure of seriation validity.

3) We plotted the number of randomly removed graves ($n$) against the proportion of seriation runs whose maximum correlation coefficient between the seriation sequence and the medians of calibrated dates falls below 0.7.

The results suggest that the solution is relatively stable: only when more than 30–40 graves are removed at random does the proportion of solutions with correlation values below 0.7 exceed 10% (Fig. 10).

Randomly removing graves from the sample can affect the seriation results to varying degrees, depending on which graves are excluded. Omitting graves that contain only a single pot is unlikely to alter the overall sequence significantly, as such graves contribute little chronological information. In contrast, removing graves with multiple pots can substantially influence the outcome, since these graves carry more

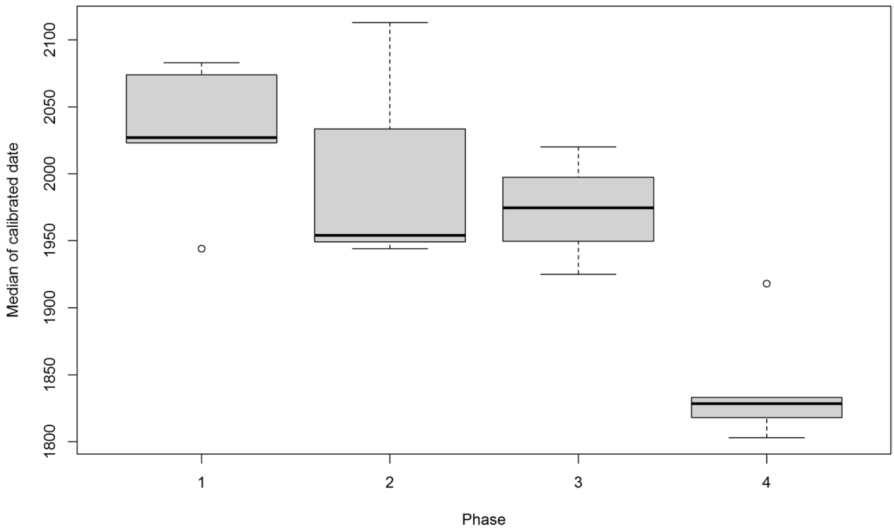


**Fig. 9** Distribution of medians of calibrated radiocarbon dates by seriation phase

information and act as key links connecting different parts of the sequence. For example, two single-pot graves are likely to appear either identical or very distant in correspondence analysis space, depending entirely on the placement of multi-pot graves that connect them through shared types. This dynamic likely explains why certain sequences change when specific graves are omitted.

Finally, we compare our proposed relative chronology of the Mokrin necropolis with the work of previous scholars. We have already demonstrated in a previous paper that some of the relative chronologies of the Mokrin necropolis were not supported by the radiocarbon dates (Krečković Gavrilović *et al.* 2025). Here, we compare our tentative chronology based on seriation to the relative chronology proposed by the excavator of the necropolis (M. Girić 1971). In his chronological framework, one-handled and two-handled pitchers were divided into three successive phases—first, second, and third—following I. Bona's (1975) relative chronology for the ceramics of the Maros cultural group. Amphorae were not subdivided into distinct phases in detail within this publication, while bowls were regarded as chronologically non-diagnostic. The one-handled pitchers—classified by M. Girić into two groups, Phase I and Phase I/II (the latter representing forms occurring in both phases)—appeared across all four phases identified in our analysis, but their frequency is higher in the first two phases (Table 1). On the other hand, the two-handled pitchers, which M. Girić divided into three groups (Phase I, Phase II, and Phase III), correspond more closely to our results. Vessels assigned to Phase I by M. Girić predominantly appeared in our Phase 1, while those attributed to Phases II and III tended to cluster within our Phases 3 and 4. So, we can conclude that the relative chronology that we propose is broadly consistent with the phasing of ceramic types proposed by M. Girić.

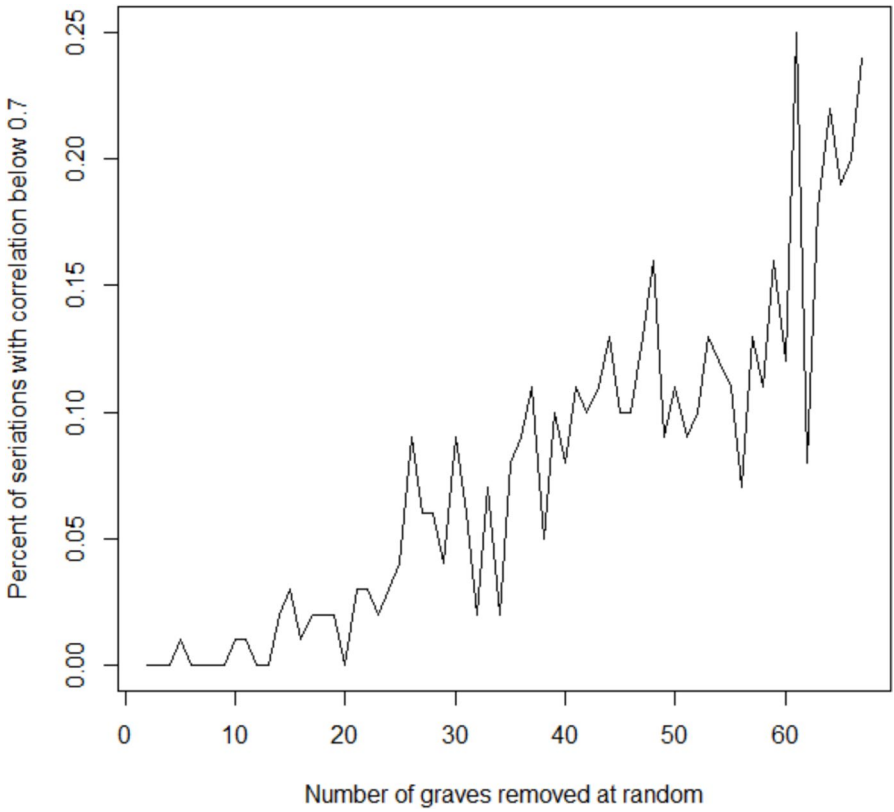


**Fig. 10** The number of graves removed at random against the correlation between the seriation sequence and absolute dates

## Conclusion

More than simply evaluating whether a particular seriation solution is consistent with absolute dates, our method actively employs radiocarbon dates to construct a valid seriation sequence by optimizing the fit between the chronological order implied by the dates and that derived from the assemblage composition. This optimization involves both the inclusion or exclusion of types and the rotation of the correspondence analysis solution to maximize the correlation.

Because the number of available radiocarbon dates is almost always much smaller than the total number of assemblages we wish to date, our approach provides a means of anchoring a large set of undated assemblages within a chronology that remains empirically grounded and statistically validated by the smaller set of radiocarbon measurements. This is a key advantage of the proposed method.

Our results suggest that the seriation of graves based on the frequencies of ceramic types, defined by vessel shapes deposited as grave goods in the Mokrin necropolis, is successful. While all the ceramic types defined for Mokrin proved

**Table 1** Comparison of Girić's (1971) relative chronology (rows) to our proposed relative chronology (columns)

| | Best solution phase 1 | Best solution phase 2 | Best solution phase 3 | Best solution phase 4 |
|---|---|---|---|---|
| One-handled pitcher Girić phase I | 4 | 8 | 3 | 4 |
| One-handled pitcher Girić phase I/II | 2 | 3 | 2 | 4 |
| Two-handled pitcher Girić phase I | 14 | 1 | 1 | 5 |
| Two-handled phase pitcher Girić II | 5 | 2 | 5 | 22 |
| Two-handled pitcher Girić phase III | 1 | 0 | 2 | 2 |

useful for constructing a relative chronology, this may not always be the case elsewhere. An important step in our approach is therefore not to assume a priori that all types are chronologically sensitive, but rather to identify explicitly those types and type combinations that are.

We propose a four-phase chronology for the Mokrin necropolis, which indicates the absence of a clear spatial pattern in its use. This finding corroborates the conclusion derived from the spatial distribution of radiocarbon-dated graves alone—that all parts of the necropolis were used contemporaneously (Krečković Gavrilović *et al.*, 2025). However, given the relatively small number of radiocarbon dates compared to the total number of graves, this four-phase relative chronology should be considered provisional and tested against additional data. Future research should therefore target graves from different phases suggested by the seriation sequence to provide an independent test of its validity. Although the number of radiocarbon dates is limited, the results presented here demonstrate the robustness of the method. We thus refrain from presenting this as a definitive relative chronology of the Mokrin necropolis, as the seriation is based on 167 out of more than 300 graves, leaving over a hundred graves out of the picture. Their chronological position can only be established through direct radiocarbon dating.

The Mokrin case study demonstrates how radiocarbon dates can be used to formally test a proposed relative chronology. By integrating seriation and radiocarbon dating in a statistically rigorous way, it provides a general model for refining archaeological chronologies beyond the Mokrin case, with potential applications to any context where assemblage data and absolute dates are present.

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**Author Contribution** All authors contributed to the research design. M.P. designed the primary method (including the R code) and performed the data analysis. M.K.G. collected and generated new radiocarbon data. M.R. and M.K.G. gathered data on pottery shapes. M.R. prepared the database, conducted the geometric morphometric analysis, and developed the associated R code. All authors contributed to writing the main manuscript and reviewed its final version.

**Data Availability** The data and code for reproducing our results are available as supplementary information files.

## Declarations

**Competing interests** The authors declare no competing interests.

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